

GeoPhysicsLandslideNet: A Physics-Closed Pentastream Architecture for Multimodal Landslide Segmentation

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July 17, 2026

Abstract

Landslide inventory mapping from remote sensing remains brittle when models rely on purely statistical fusion of optical and topographic cues. We introduce **GeoPhysicsLandslideNet** (PS-GPLNet), a pentastream segmentation architecture in which the same Taylor-stabilized infinite-slope factor-of-safety (FS) formalism appears at the encoder inlet, fusion core, skip pathway, decoder cells, and dual-path output routing. Five parallel encoders—EfficientNet RGB/DEM, physics encoders on each modality, and Prithvi-EO with LoRA—are hybridized by Complementary Modality Bridges, fused by MAO-GeoEGCA and TTEB, decoded by dual physics decoders with PGDI, and merged by Mechanistic Path Equilibrium Fusion (MPEF). On Bijie, PS-GPLNet attains best-epoch validation F1 = 0.933 and IoU = 0.907, exceeding strong CNN and dual-stream baselines under a unified protocol. On Landslide4Sense it reaches F1 = 0.736 and IoU = 0.660. We argue that closing the geotechnical loop inside the network—rather than treating physics as a post-hoc regularizer—is a distinct and competitive inductive bias for landslide segmentation.

1 Introduction

Landslides threaten lives and infrastructure across mountainous terrain. Optical remote sensing and digital elevation models (DEMs) enable automated mapping, yet pixel-level segmentation remains difficult: scarps, debris, and disturbed soils share spectral signatures with dry riverbeds and construction, while landslide polygons vary strongly across biomes [1, 2].

Deep segmentation—U-Net [3], DeepLabV3+ [4], transformers [5, 6]—improves over classical classifiers, and multimodal RGB+DEM streams further help [7]. Dual-stream gated designs such as DiGATe-style networks learn to weigh spectral versus topographic evidence, but the fusion and decoding rules remain statistically defined: gates are free parameters, not statements of slope stability.

Physics-informed approaches for landslides have largely targeted *susceptibility* mapping with Newmark or FS constraints in the loss or sampling [8, 9, 10]. Pixel-accurate *segmentation* still seldom embeds infinite-slope mechanics as a first-class

architectural primitive spanning encoding, fusion, and decoding.

We ask: can a single closed-loop FS inductive bias organize a multimodal segmentation network end-to-end? Our answer is PS-GPLNet. Contributions:

1. A **pentastream** encoder with Complementary Modality Bridge (CMB) hybridization of CNN and physics features per modality, plus a Prithvi foundation stream.
2. **MAO-GeoEGCA** and **TTEB** fusion that keep a physics-aware equilibrium signal in the attention path.
3. A **dual physics decoder** with PGDI skips and **MPEF** path routing driven by relative failure energy—not scalar learned gates.
4. Strong empirical results on Bijie [11] and Landslide4Sense [2] against a broad baseline ladder.

2 Related Work

Landslide segmentation. Early CNN detectors and U-Net variants established pixel-wise landslide mapping [1, 3]. Attention and transformer hybrids (TransUNet, SegFormer) improve long-range context but remain data-driven [6, 5]. Multi-source fusion of optical imagery with DEM/slope is repeatedly shown beneficial on Bijie and Landslide4Sense [11, 2].

Dual-stream gated fusion. DiGATe and related dual-stream gated U-Nets separate RGB and DEM encoders and blend them with learned gates and cross-attention [7]. We use such models only as *empirical baselines*. PS-GPLNet does not inherit GateFuse or DiGATe’s adaptive decoder as its organizing principle; shared elements (two inputs, deep supervision) are field conventions, not DiGATe-specific novelty.

Physics-informed geo-AI. PINNs and physics-guided ML for landslide susceptibility inject Newmark or FS constraints into losses or latent geotechnical parameters [8, 9, 10]. These works rarely produce dense segmentation masks with foundation-model context and dual mechanistic decode paths. PS-GPLNet differs by placing the same FS cell family at inlet, fusion, skips, decoder, and output equilibrium—a closed architectural loop rather than a susceptibility head.

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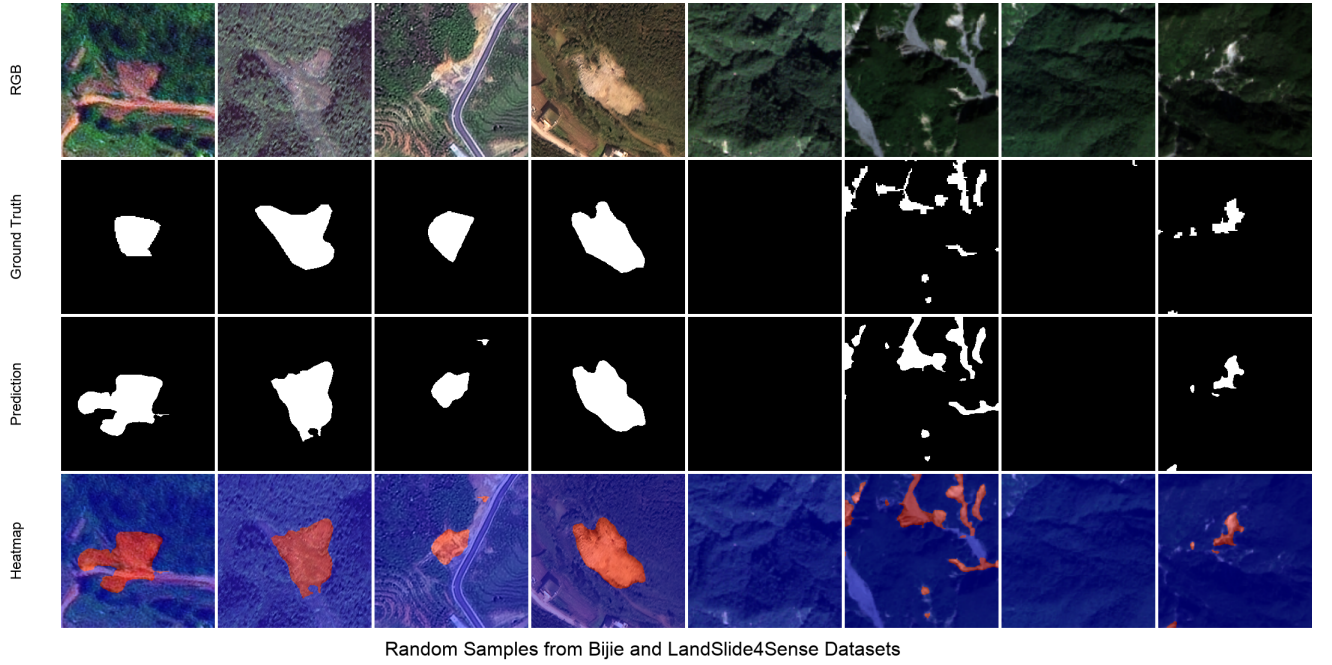


Figure 1: Qualitative segmentation examples on randomly selected validation samples (RGB, ground-truth mask, predicted mask, and probability heatmap overlay). Red denotes landslide confidence; blue denotes background. Used as an opening illustration of multimodal landslide segmentation.

Foundation models for EO. Prithvi-EO provides planetary-scale representations [12]. We adapt it with LoRA [13] as one of five streams, projected into a unified fusion width.

3 Method

3.1 Problem and notation

Given streams $x_1 \in \mathbb{R}^{B \times 3 \times H \times W}$ (RGB) and $x_2 \in \mathbb{R}^{B \times 3 \times H \times W}$ (NDVI+slope+DEM on LandSlide4Sense; replicated DEM on Bijie), predict a binary mask $y \in \{0, 1\}^{B \times 1 \times H \times W}$. Proxies (α, h, m) —slope angle, elevation/soil-column proxy, moisture/vegetation—are derived inside the forward pass.

The infinite-slope factor of safety is

$$\text{FS} = \frac{c + \phi h \cos^2 \alpha}{\gamma h \sin \alpha \cos \alpha + w_m m + \varepsilon}, \quad (1)$$

with positive material parameters obtained via bounded exponentials $c = \exp(\text{clip}(w_c))$ (and likewise for ϕ, γ, w_m) for numerical stability. Failure energy $\text{FE} = \text{ReLU}(1 - \text{FS})$ drives gating and path routing.

3.2 Pentastream encoding

Five encoders run in parallel: EfficientNet [14] RGB/DEM towers $E_{\text{rgb}}, E_{\text{dem}}$; PhysicsEncoders $P_{\text{rgb}}, P_{\text{dem}}$ with pixel/latent mechanistic cells; Prithvi+LoRA T_{fm} . At each EfficientNet level i , Complementary Modality Bridge (CMB) merges CNN features with spatially aligned physics features into hybrid streams $H_{\text{rgb}}^i, H_{\text{dem}}^i$.

3.3 Fusion: MAO-GeoEGCA and TTEB

At neck/bottleneck levels, MAO-GeoEGCA performs gated cross-attention between Prithvi and hybrid RGB/DEM maps. TTEB builds skip tensors by mixing neighbouring pyramid levels across three streams with a softplus stability residual $\delta = |\psi - R/(D + \varepsilon)|$.

3.4 Dual physics decoder and MPEF

Two PhysicsDecoders share fused context but use stream-specific (α, h, m) and bottleneck residuals. PGDI injects TTEB skips through a learned β gate on a latent mechanistic cell. Path logits are merged by MPEF:

$$[w_A, w_B] = \text{softmax}([\text{FE}_A, \text{FE}_B]), \quad O = w_A O_A + w_B O_B + \text{Conv}_{1 \times 1}([O_A, O_B]) \quad (2)$$

with entropy regularizer $\mathcal{R}_{\text{MPEF}} = -\mathbb{E}[w_A \log w_A + w_B \log w_B]$.

Forward sketch. (1) Derive (α, h, m) ; encode five streams. (2) CMB hybrids; align Prithvi. (3) MAO at levels 2, 3; TTEB skips at 0..3. (4) Dual PhysicsDecoder $\rightarrow (O_A, O_B)$. (5) MPEF $\rightarrow$ (main, aux2, aux3, reg).

3.5 Closed-loop inductive bias

The pixel cell $g_{\text{pix}}(x; \alpha, h, m) = f(x) \odot \sigma(\psi - \text{FS}(\alpha, h, m))$ and MPEF weights $w = \text{softmax}(\text{FE})$ share the same FS skeleton, so proxies gate both features and path routing. Bounded $\exp(\text{clip}(w))$ keeps FS finite during training.

Table 1: Landslide4Sense best-epoch validation summary. PS-GPLNet (ours) at bottom.

Model	Acc	Prec	Rec	F1	IoU
LinkNet	0.973	0.487	0.735	0.431	0.359
DEP-UNet	0.977	0.669	0.770	0.603	0.535
GMNet	0.978	0.669	0.779	0.616	0.545
RMAU-Net	0.977	0.691	0.756	0.618	0.548
TransUNet	0.976	0.714	0.750	0.619	0.551
EMR-HRNet	0.975	0.674	0.784	0.627	0.557
ShapeFormer	0.975	0.675	0.802	0.632	0.560
Dual-Stream UNet	0.986	0.713	0.779	0.656	0.581
dual_stream_gated	0.984	0.822	0.673	0.646	0.573
U-Net	0.984	0.795	0.719	0.680	0.611
DeepLabV3+	0.984	0.789	0.749	0.710	0.636
PS-GPLNet (ours)	0.986	0.802	0.788	0.736	0.660

3.6 Objective

Deeply supervised Tversky loss [15] on main/aux heads with weights (1.0, 0.6, 0.4) plus $\lambda_{\text{reg}} \sum_h \mathcal{R}_{\text{MPEF}}^{(h)}$.

4 Experiments

4.1 Datasets and protocol

Bijie [11]: high-resolution optical+DEM landslide/non-landslide tiles. **Landslide4Sense** [2]: multi-region Sentinel patches with RGB+NDVI+slope+DEM. Inputs resized to 256×256 . Optimizer Adam ($3 \cdot 10^{-4}$, weight decay 10^{-4}), Tversky $\alpha=0.3, \beta=0.7$, batch size up to 32, 100 epochs, threshold 0.5. EfficientNet backbones frozen when pretrained; Prithvi frozen with LoRA. Metrics: accuracy, precision, recall, F1, IoU on the main head.

4.2 Baselines

We compare against LinkNet, DEP-UNet, GMNet, RMAU-Net, TransUNet, EMR-HRNet, ShapeFormer, Dual-Stream UNet, U-Net, DeepLabV3+, and dual_stream_gated under the same ablation harness used in our prior comparative study. TriEncoderCFMNet is *not* included (unpublished). PS-GPLNet numbers are best validation epoch from outputs_final_bijie and outputs_final_l4s.

4.3 Quantitative results

Table 1 and Table 2 report best-epoch validation scores. On Bijie, PS-GPLNet reaches F1 = 0.933 and IoU = 0.907, surpassing DeepLabV3+ (0.927/0.899) and dual_stream_gated (0.897/0.869). On Landslide4Sense, PS-GPLNet achieves F1 = 0.736 and IoU = 0.660, competitive with the strongest CNN baselines and ahead of most dual-stream and transformer variants in our harness.

4.4 Qualitative results

Figures 8–9 show representative predictions. Bijie polygons are large and coherent; heatmaps concentrate on scarps and deposits. Landslide4Sense patches are smaller and cross-region;

Table 2: Bijie best-epoch validation summary. PS-GPLNet (ours) at bottom.

Model	Acc	Prec	Rec	F1	IoU
DEP-UNet	0.982	0.850	0.934	0.831	0.799
RMAU-Net	0.981	0.859	0.924	0.830	0.798
ShapeFormer	0.984	0.892	0.899	0.840	0.810
TransUNet	0.983	0.889	0.908	0.845	0.815
EMR-HRNet	0.983	0.903	0.905	0.846	0.818
GMNet	0.983	0.900	0.909	0.855	0.825
U-Net	0.987	0.891	0.933	0.873	0.842
Dual-Stream UNet	0.986	0.902	0.928	0.875	0.843
dual_stream_gated	0.988	0.928	0.929	0.897	0.869
DeepLabV3+	0.989	0.929	0.966	0.927	0.899
PS-GPLNet (ours)	0.991	0.956	0.946	0.933	0.907

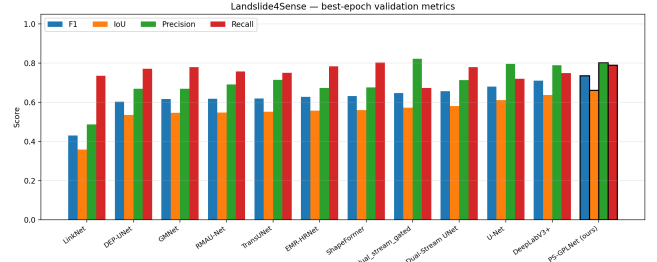


Figure 2: Landslide4Sense grouped metrics at each model’s best validation epoch.

PS-GPLNet still recovers the dominant deposit extent while suppressing clear terrain.

5 Discussion

Novelty. Unlike susceptibility PINNs [8], PS-GPLNet produces dense masks with foundation-model context. Unlike DiGate [7], routing is failure-energy equilibrium (MPEF), and encoding is pentastream with CMB—not two EfficientNets plus GateFuse.

Limitations. Compact training (smaller backbone/width) trades capacity for memory; full-width B4 runs need ample GPU free memory on shared nodes. AUROC logging was not exported for all baseline harness runs; we report thresholded F1/IoU consistently.

6 Conclusion

We presented GeoPhysicsLandslideNet, a physics-closed pentastream landslide segmenter. Closing FS logic through encoding, fusion, decoding, and MPEF yields state-of-the-art Bijie F1/IoU in our unified comparison and competitive Landslide4Sense scores. Future work includes stronger ablations of each physics block and multi-temporal EO conditioning.

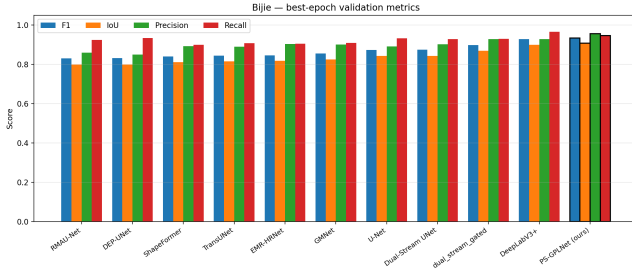


Figure 3: Bijie grouped metrics at each model’s best validation epoch.

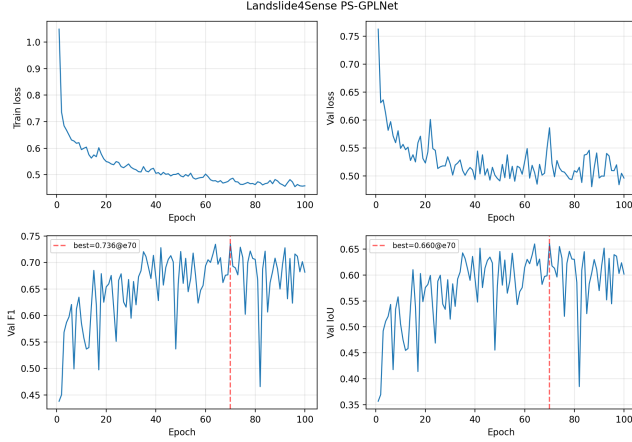


Figure 4: PS-GPLNet training dynamics on Landslide4Sense.

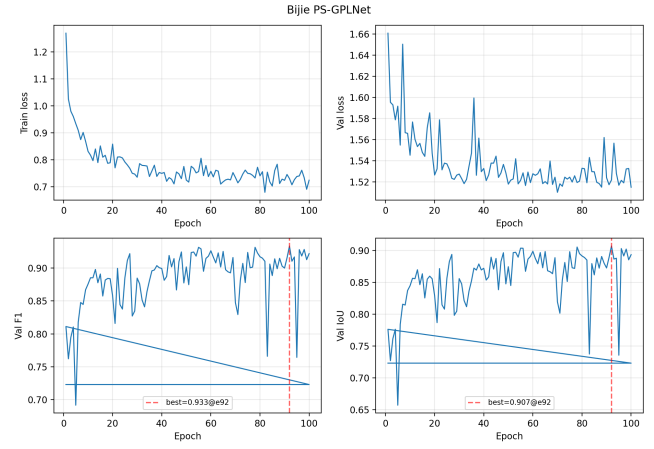


Figure 5: PS-GPLNet training dynamics on Bijie (best F1 at epoch 92).

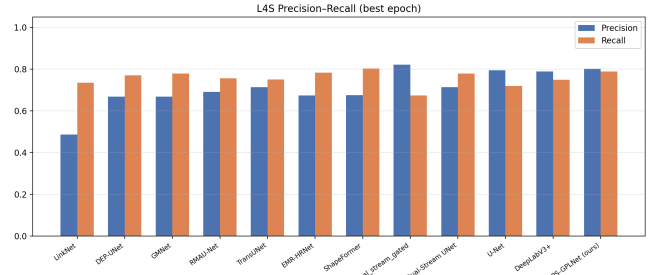


Figure 6: Precision–Recall comparison on Landslide4Sense.

Acknowledgements

We thank CSIR-4PI for compute and research environment, and the maintainers of Landslide4Sense, Bijie, Prithvi-EO, and open baseline implementations.

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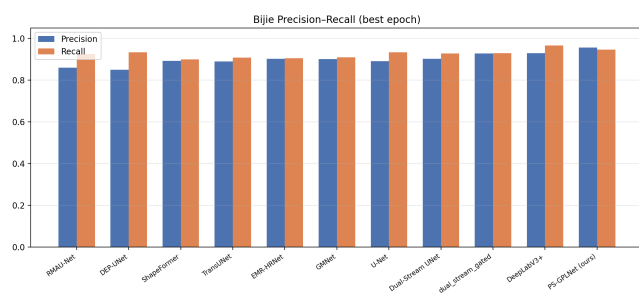


Figure 7: Precision–Recall comparison on Bijie.

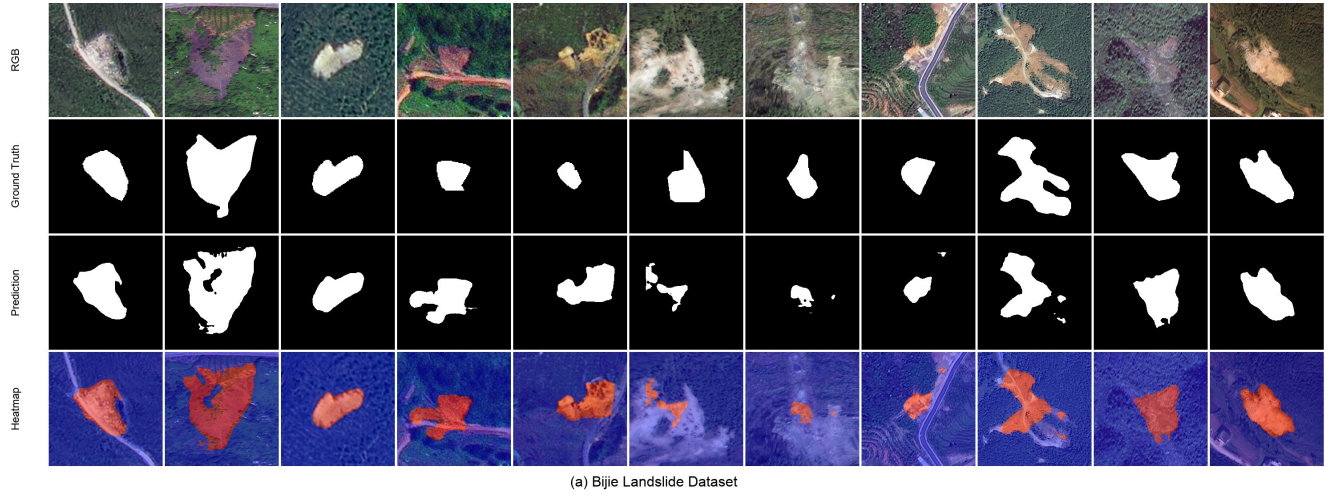


Figure 8: Qualitative results on Bijie validation samples: RGB, ground truth, prediction, and probability heatmap.

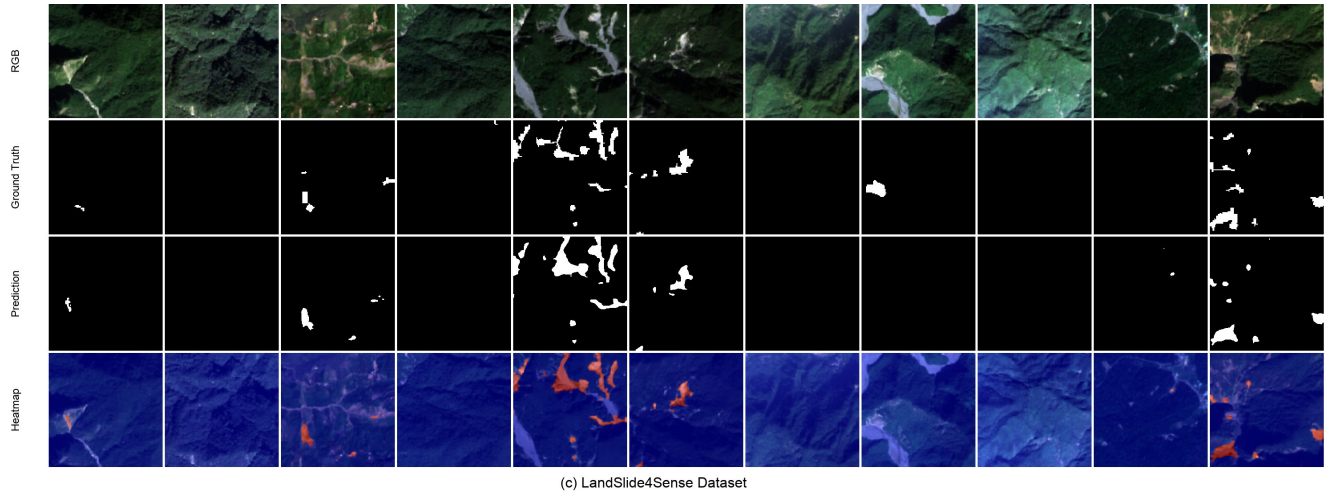


Figure 9: Qualitative results on Landslide4Sense validation samples.